



BITS Pilani

Automotive Vehicle

29-08-2026

Dr. Srinibas Tripathy
Department of Mechanical Engineering



Automotive Vehicle [AELZC441]

Lecture No. 06: Fundamentals of IC Engines

Recap of previous section



- ① vehicle propulsion sys
- ② Intake.
- ③ ~~Ch~~ vehicle structure - chassis & body

Sections to be covered

Contact Session	List of Topic Title	Sub-Topics
1	Introduction to automobiles	- Automobiles, classification, and homologation - Development methodology, virtual engineering methods
2	Chassis and Body	- Types of chassis and applications - Weight distribution and analysis
3	Powertrain Fundamentals	- Vehicle propulsion system requirements - Tractive force and torque requirement - Powertrain architecture
4,5,6	IC Engine Fundamentals	- Fundamentals of IC Engine - Fuel conversion techniques - Torque characteristics - Emissions in engines

IC Engine Fundamentals



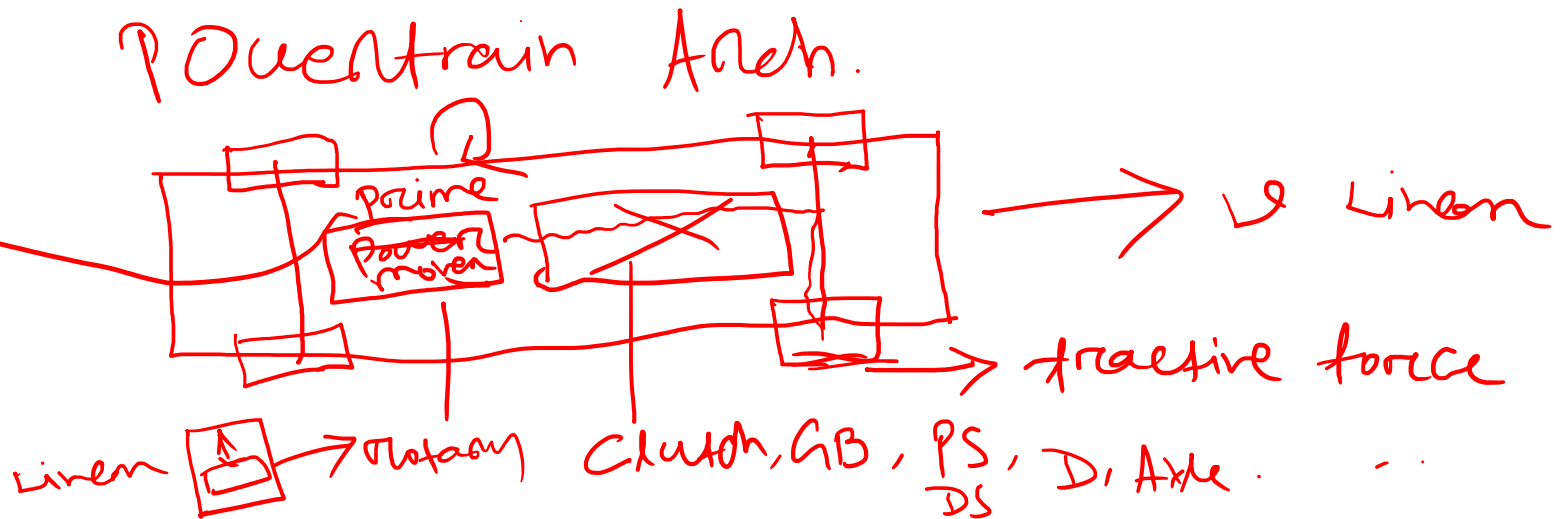
- Fundamentals of IC Engine
- Fuel conversion techniques
- Torque characteristics
- Emissions in engines

Prime movers

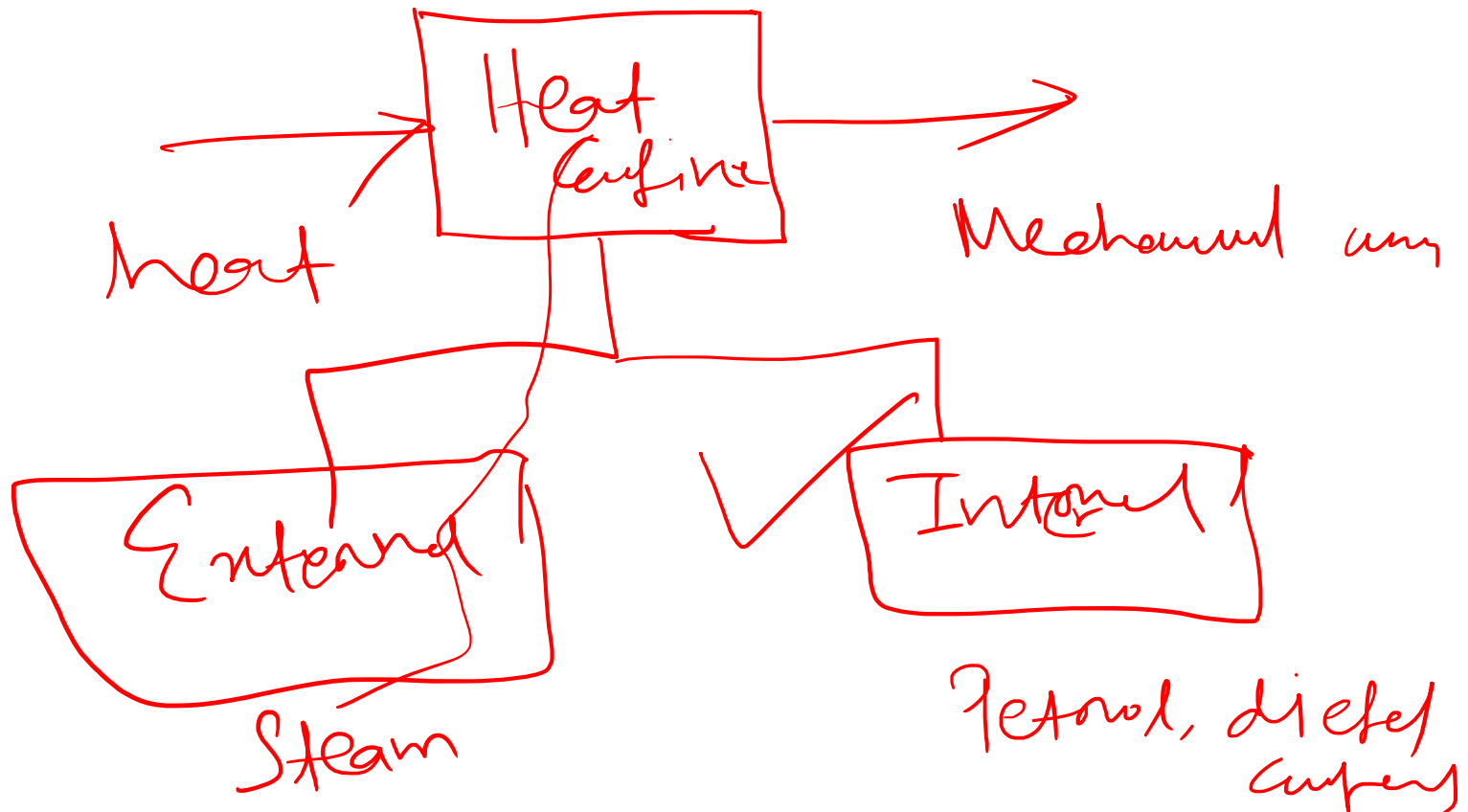


- ❖ Stored energy to power: Mechanical rotation and vehicle propulsion.

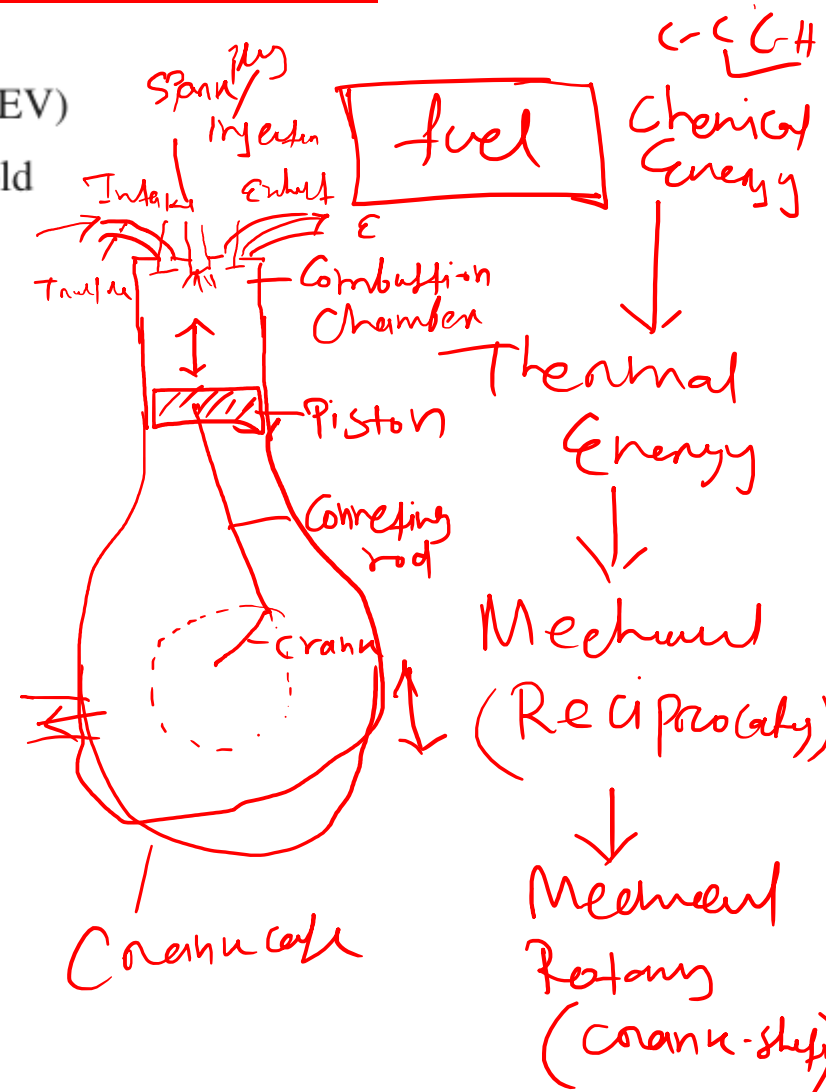
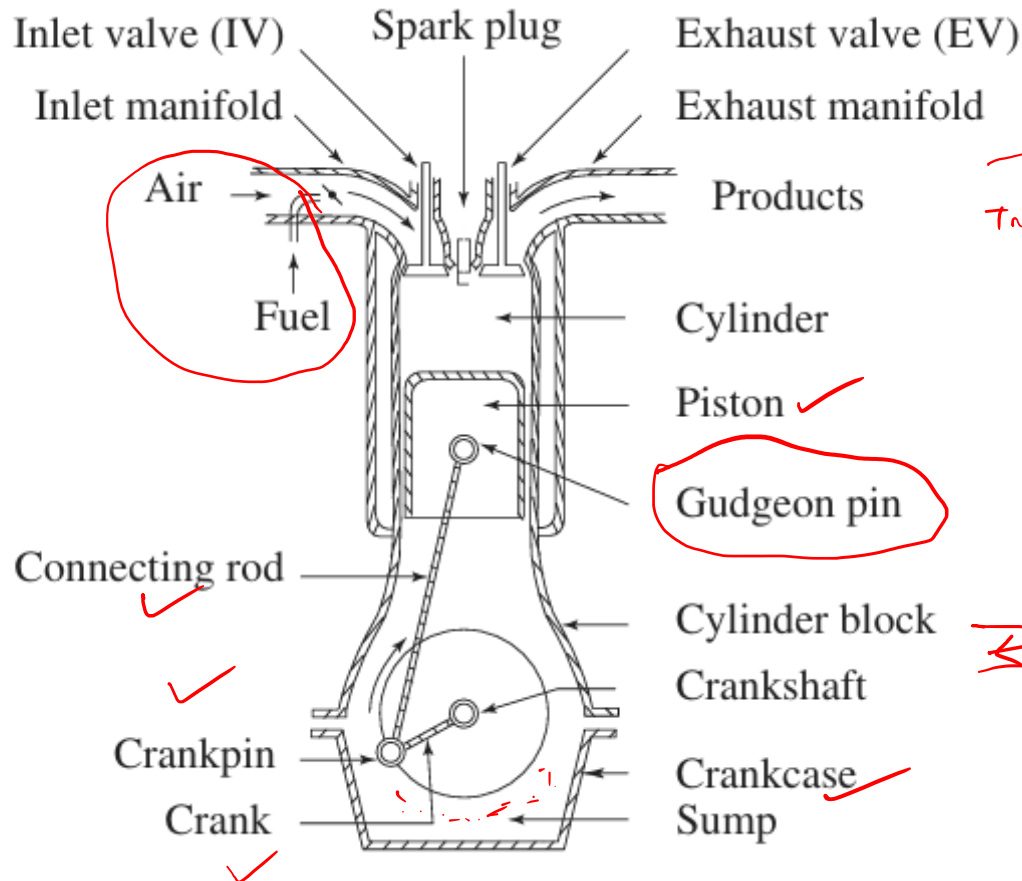
Prime Mover	Energy Source	Energy Conversion
Internal Combustion Engine	Petrol, CNG, H ₂ , Diesel	Chemical to Thermal to Mechanical
Electric Motor	Battery	Electrical to Mechanical
Fuel-Cell + Motor	Hydrogen	Chemical to Electrical to Mechanical
Hybrid Powertrain	Fuel + Battery	Chemical and Electrical to Mechanical



Internal Combustion Engine



Spark-Ignition Engine: Components



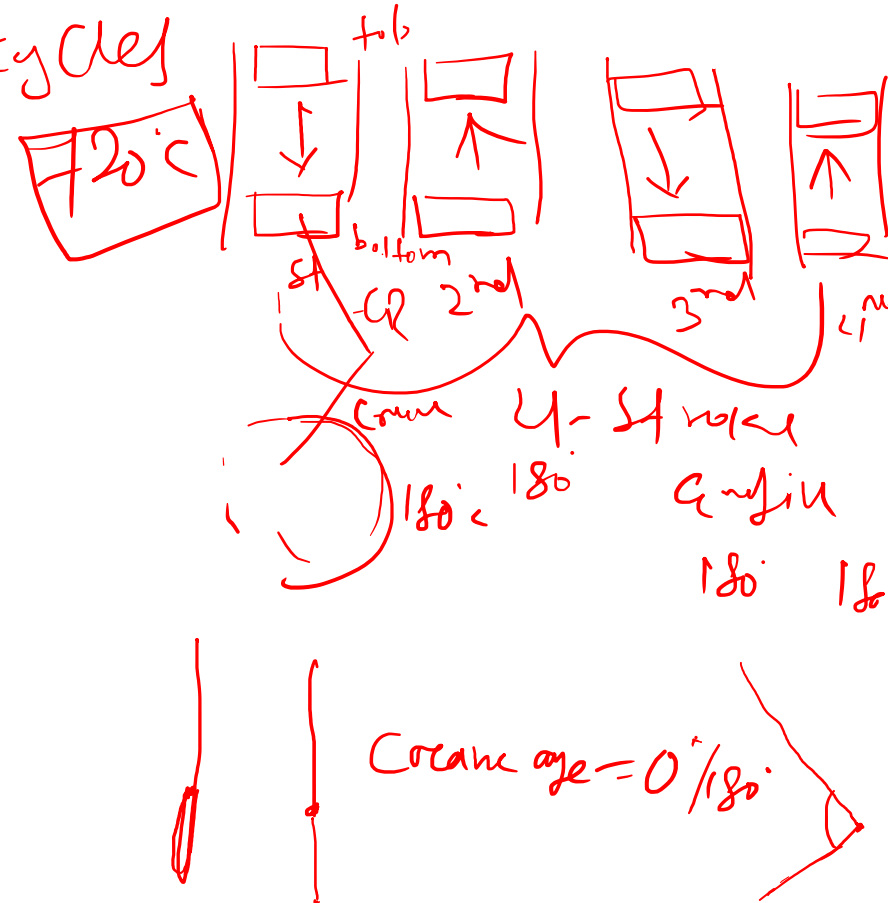
IC Engine: Classification

1. Working Cycle

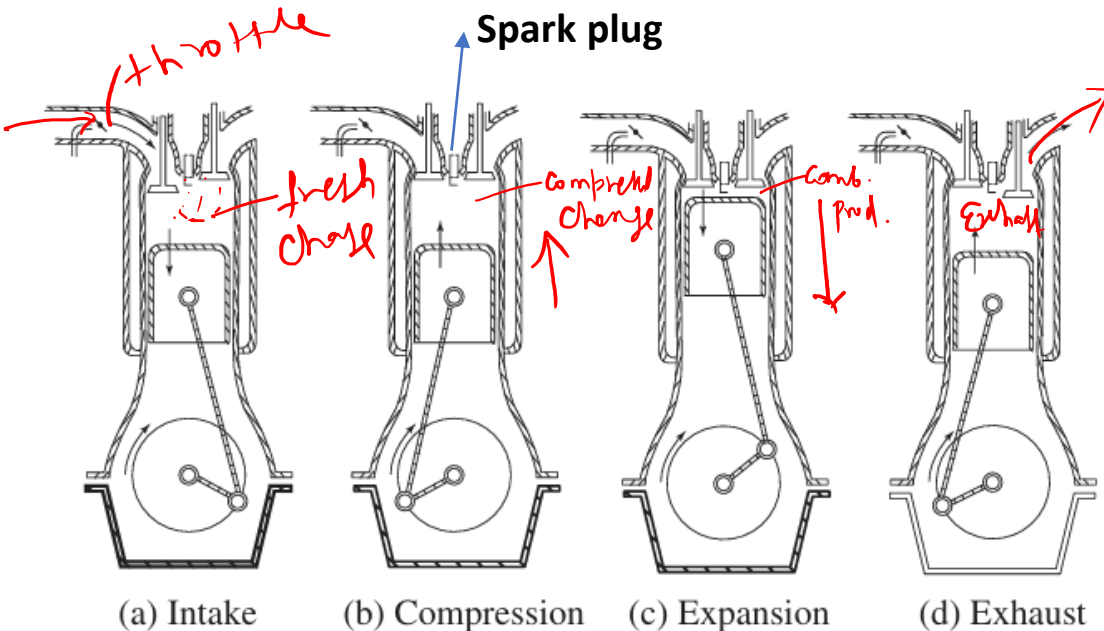
- Four-stroke cycle engines: ✓ → 2-cycles
- Two-stroke cycle engines: ✓

2. Method of Ignition

- Spark ignition (SI) ✓
- Compression ignition (CI) ✓

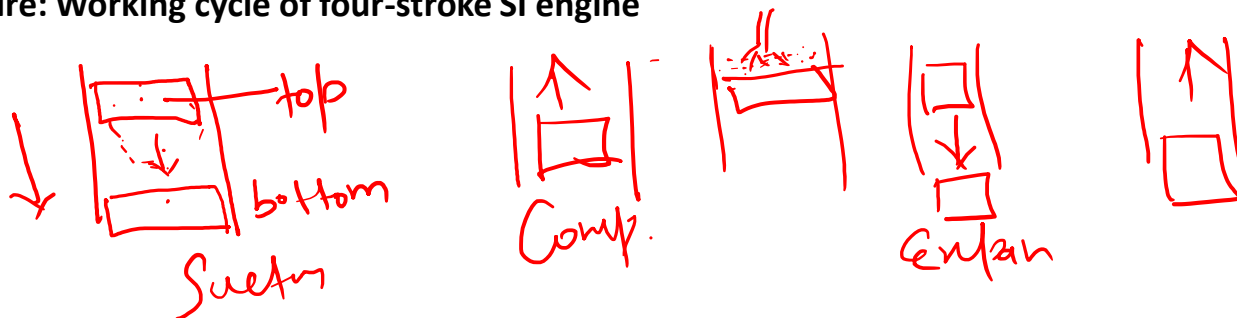
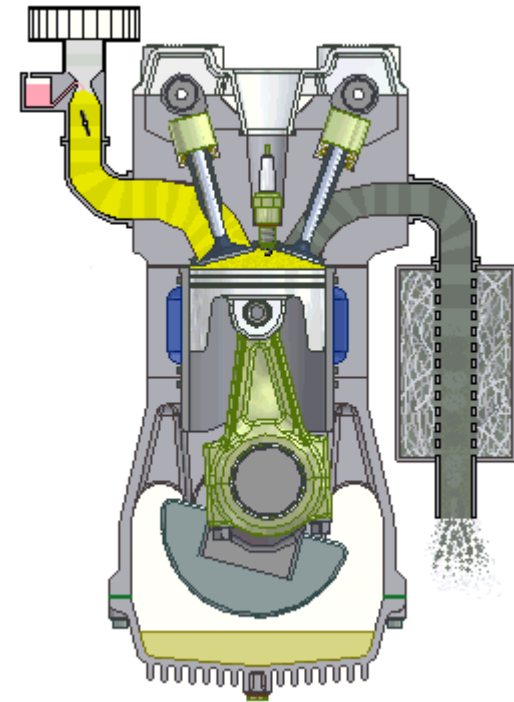


Four-stroke Spark-Ignition Engine: Working

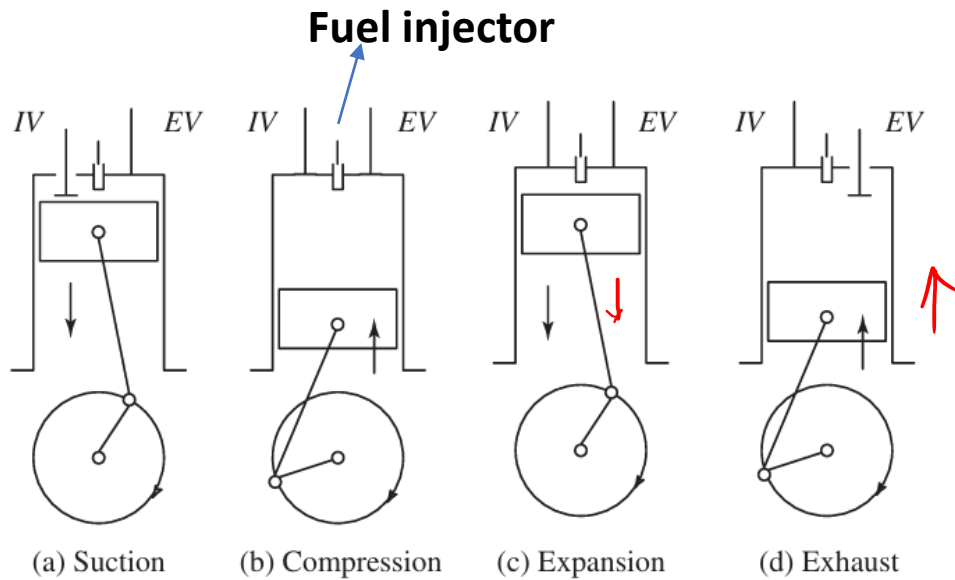


Suction

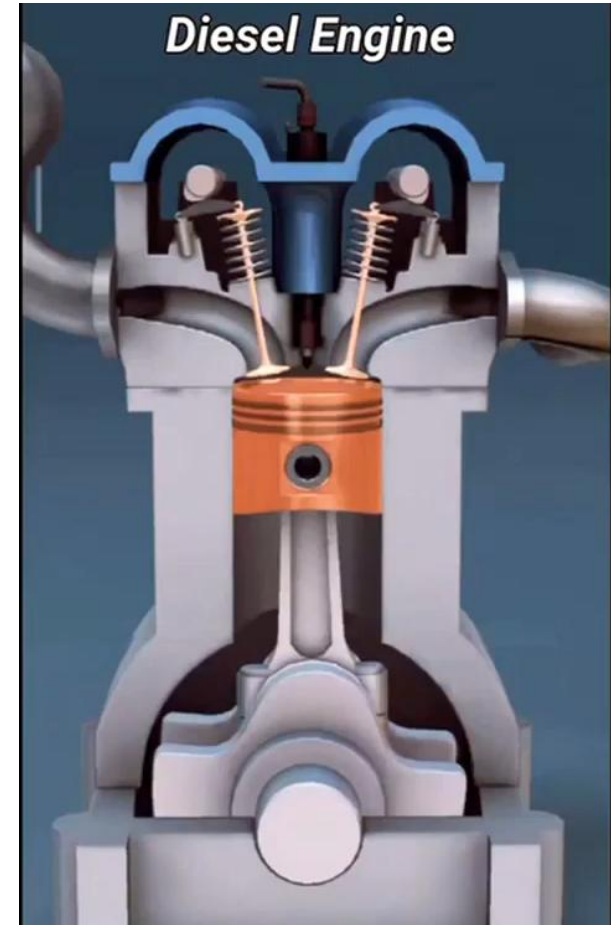
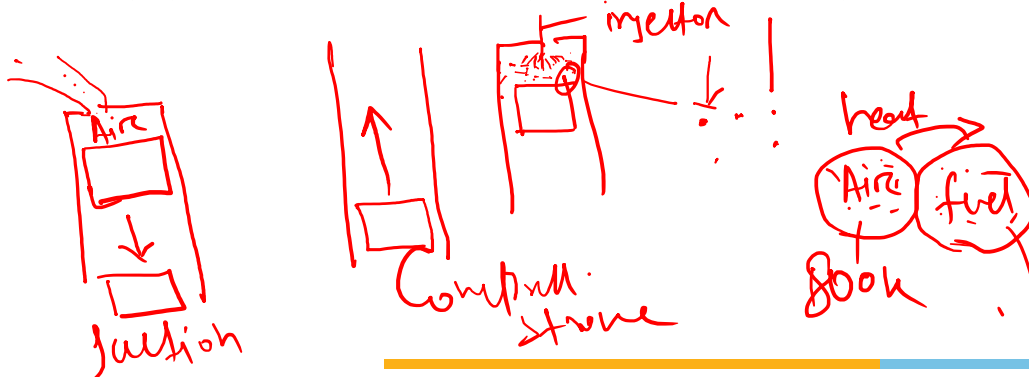
Figure: Working cycle of four-stroke SI engine



Four-Stroke Compression-Ignition Engine: Working



Working cycle of four-stroke Compression-Ignition engine



Two-Stroke Engine: Working

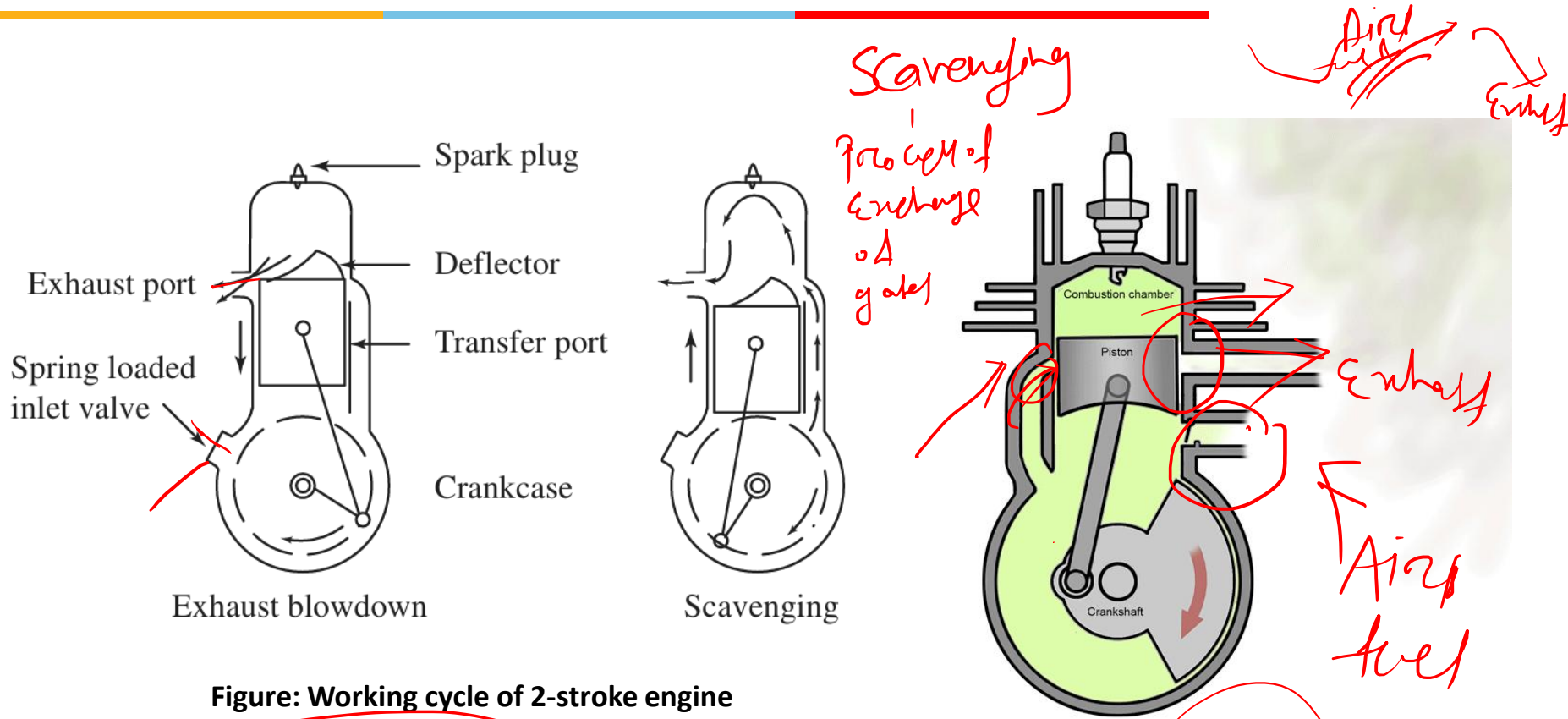
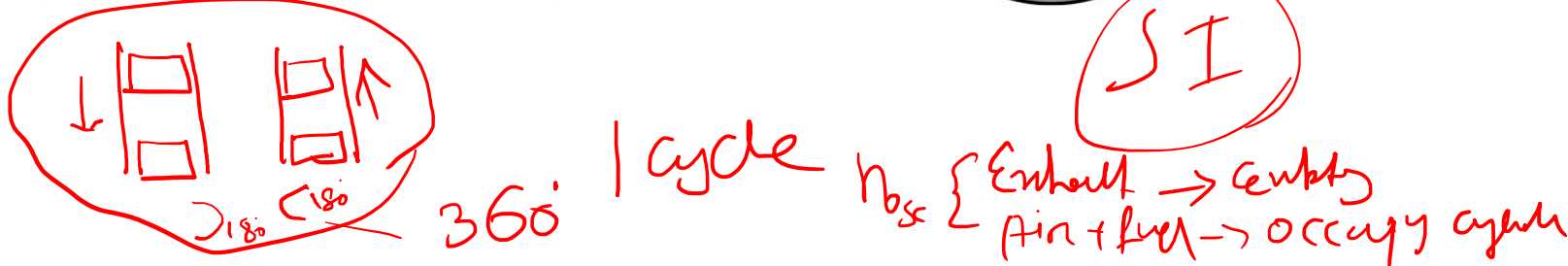


Figure: Working cycle of 2-stroke engine



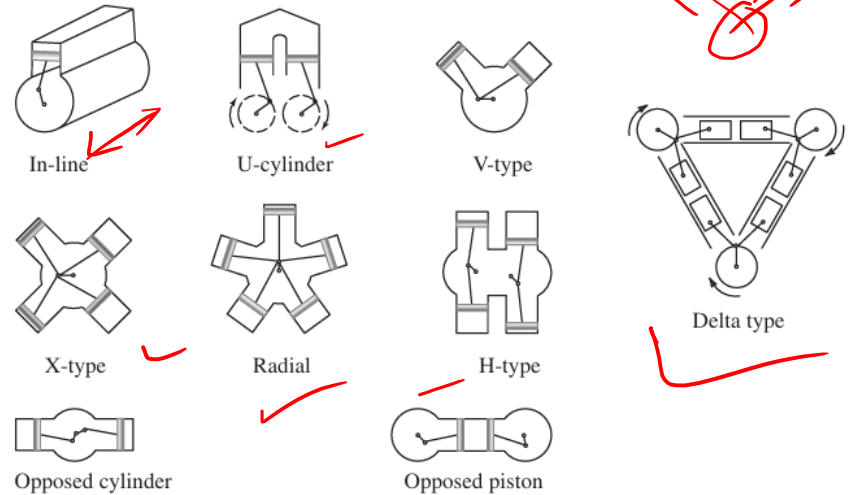
IC Engine: Classification



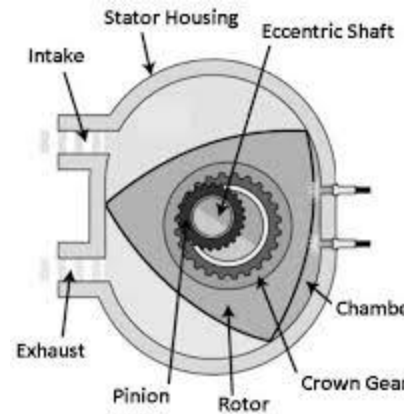
Based on design ✓

- **Reciprocating engines**

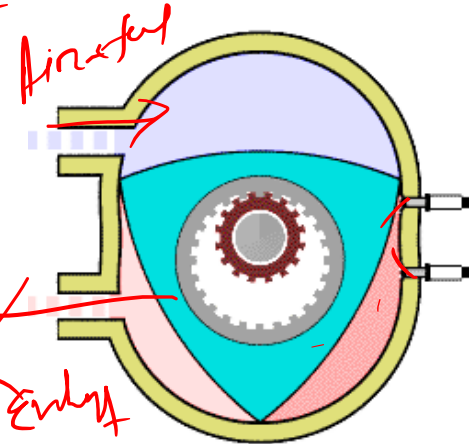
In-line
V-type
Radial
Horizontally opposed



- **Rotary engines**
Wankel engine



Wankel Rotary Engine



Spark-Ignition Engine: Components

Cams

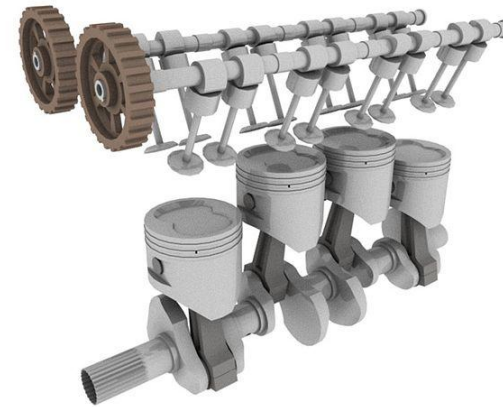
Cams are projections on the camshaft that actuate the valves. They ensure correct valve timing and duration.

→ swirl
→ Tumble
→ Squish

Clp
Section
TDC

Cam
Section

Crankshaft

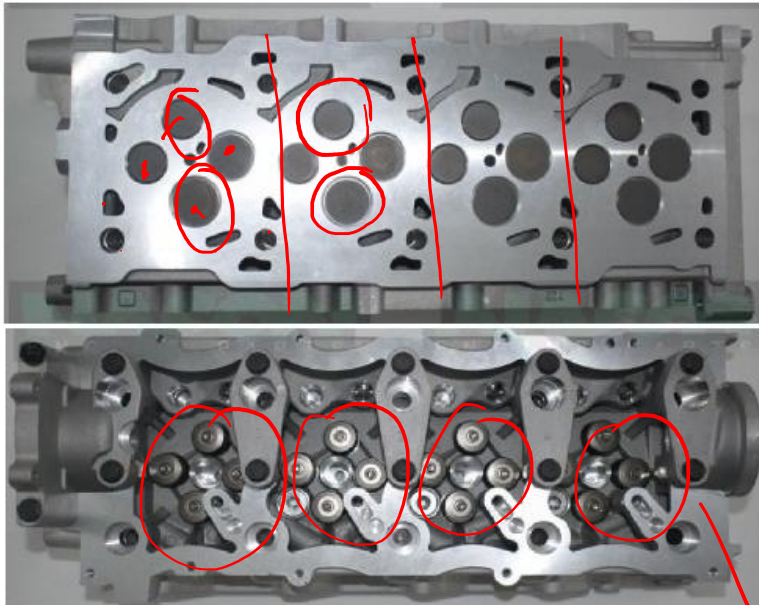


piston

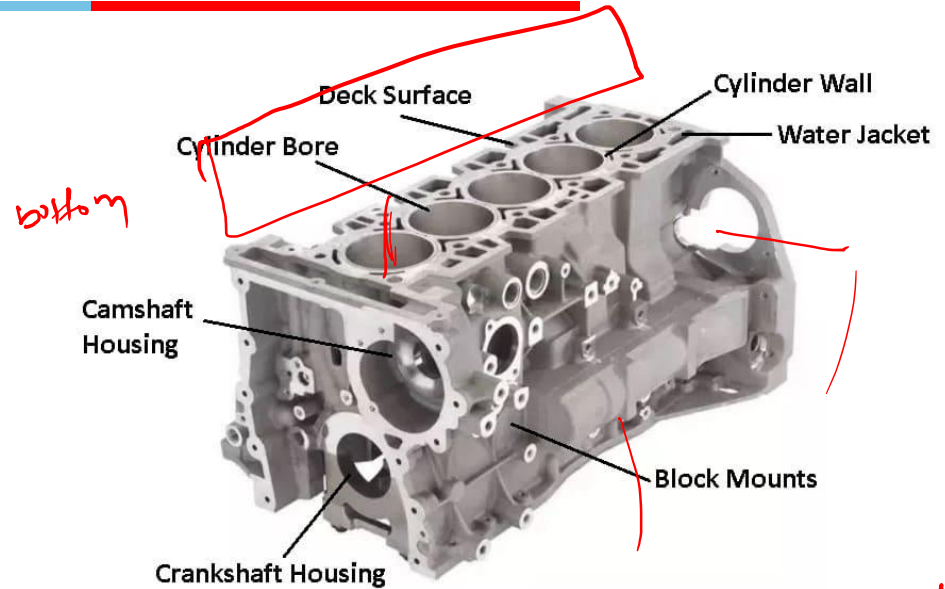
Crankshaft
Clutch
AB
Fly wheel

C.R

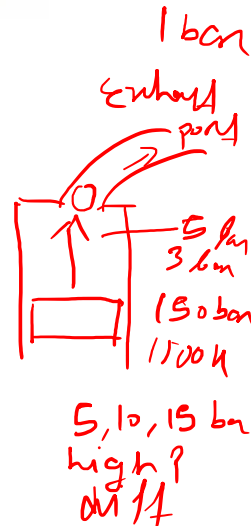
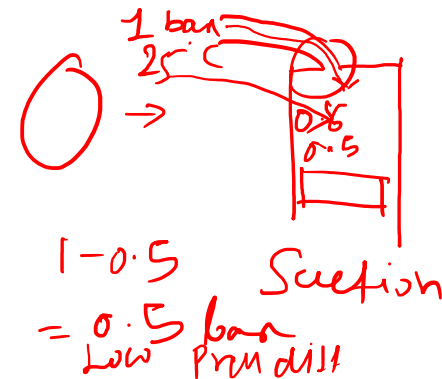
Spark-Ignition Engine: Components



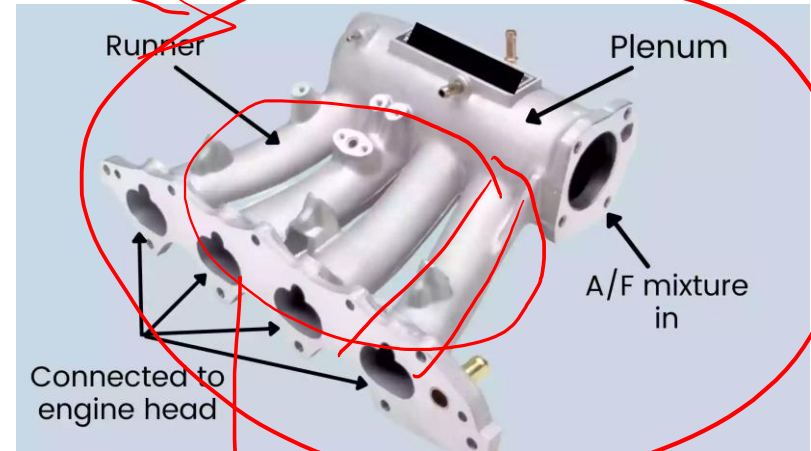
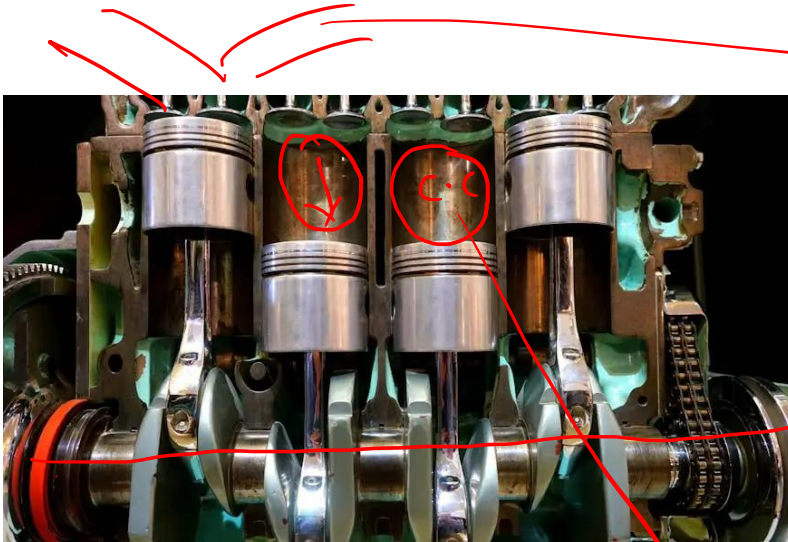
Cylinder head



Cylinder block

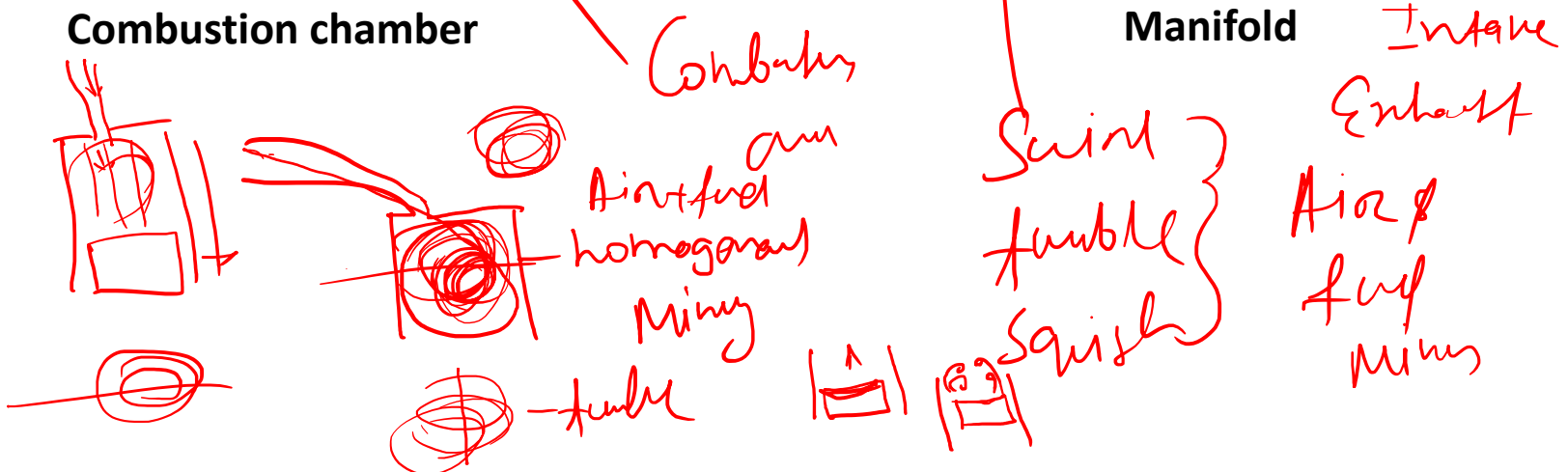


Spark-Ignition Engine: Components

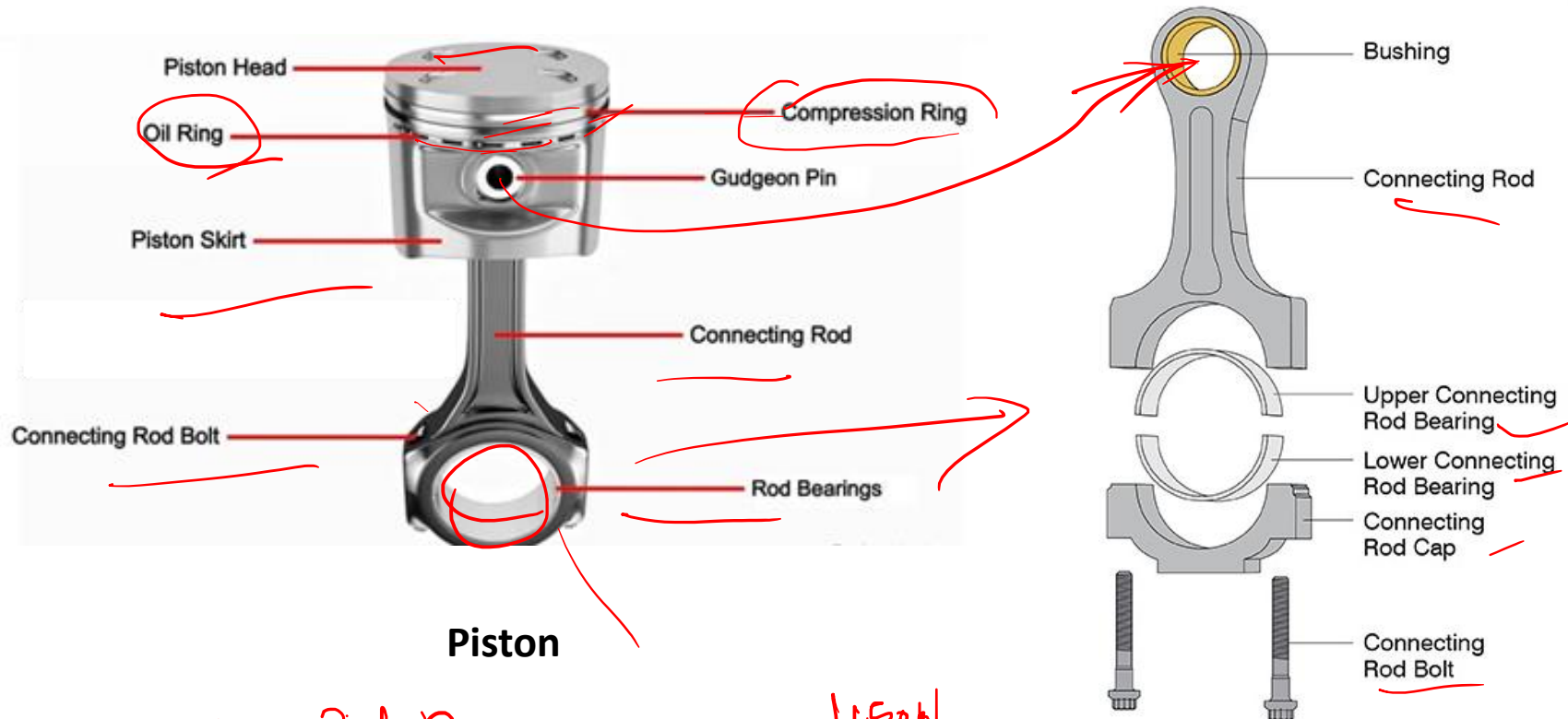


Combustion chamber

Manifold



Spark-Ignition Engine: Components



Linear Piston

blow-by crankcase

Connecting rod

Spark-Ignition Engine: Components



Piston Rings

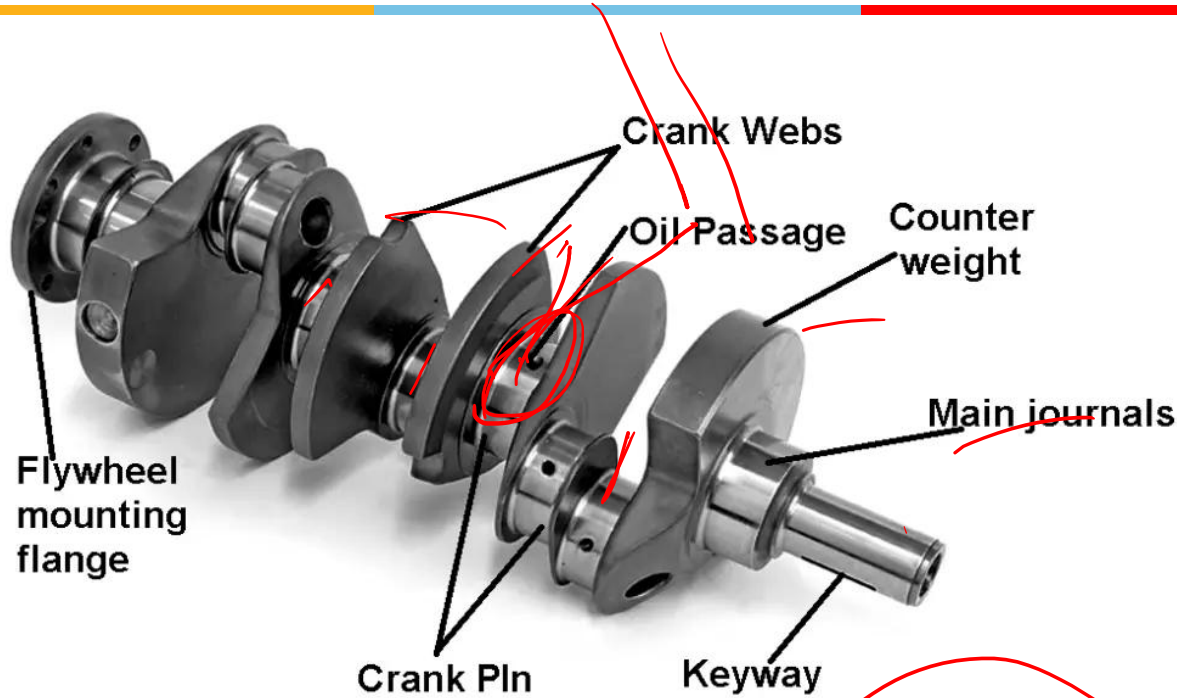
Piston rings provide a gas-tight seal between the piston and the cylinder wall.

Gudgeon Pin

The gudgeon pin connects the piston to the small end of the connecting rod. It allows relative motion between them.



Spark-Ignition Engine: Components



① 3-2-4
↗ ↘ ↗

Camshaft

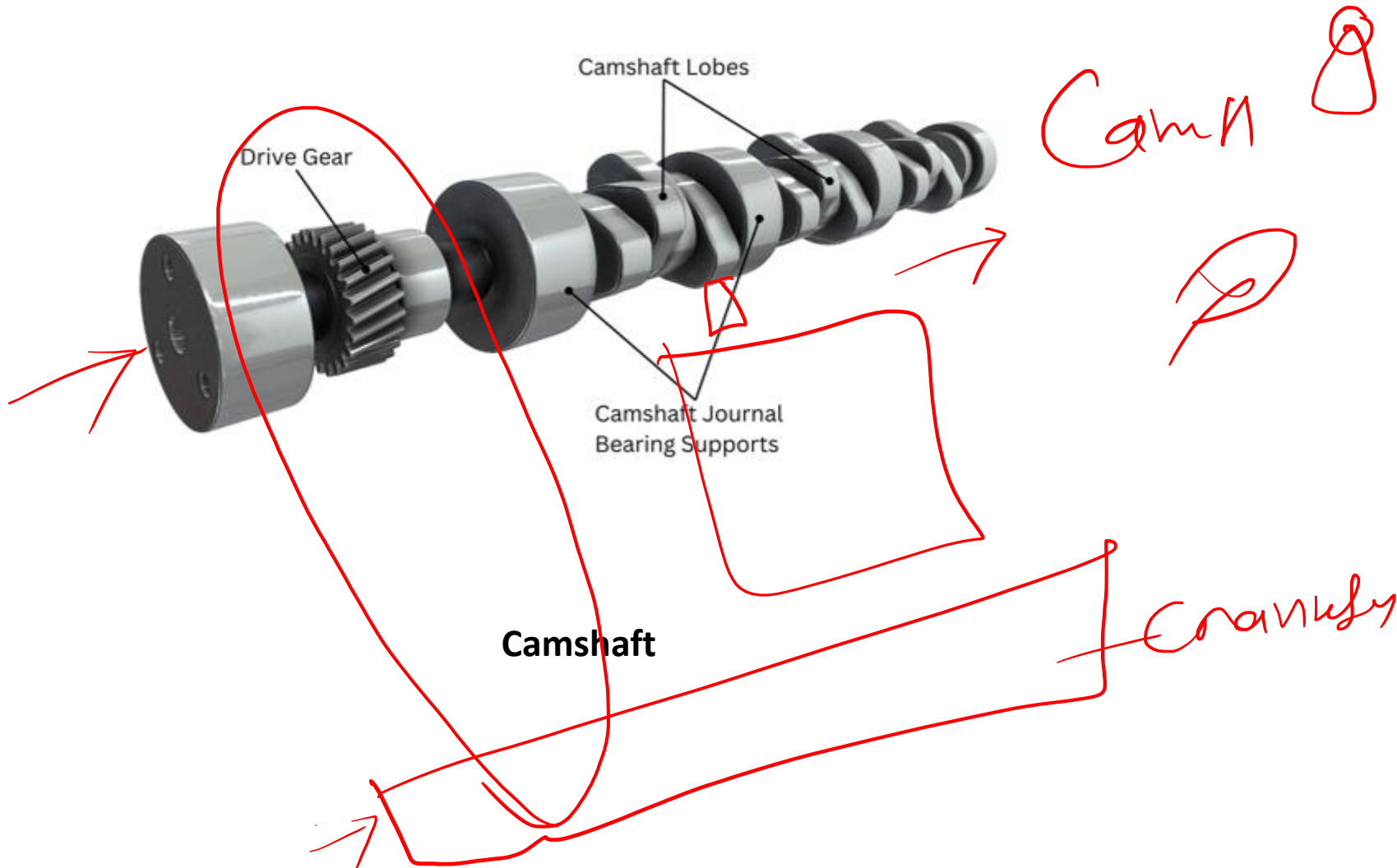
Assembly

Crankshaft

giving space at different times



Spark-Ignition Engine: Components



Nomenclature of SI Engine



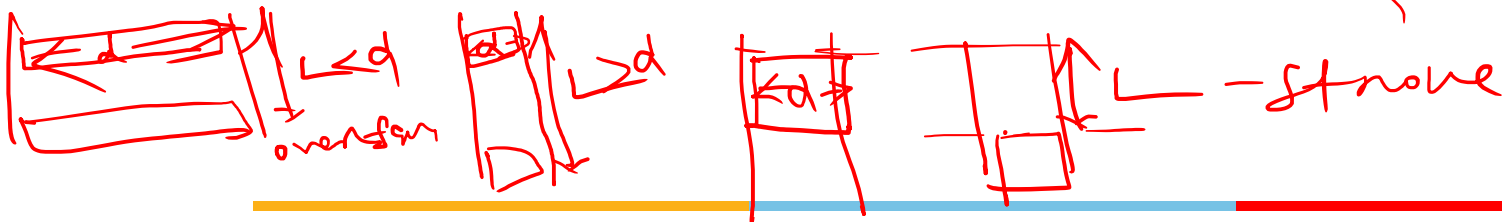
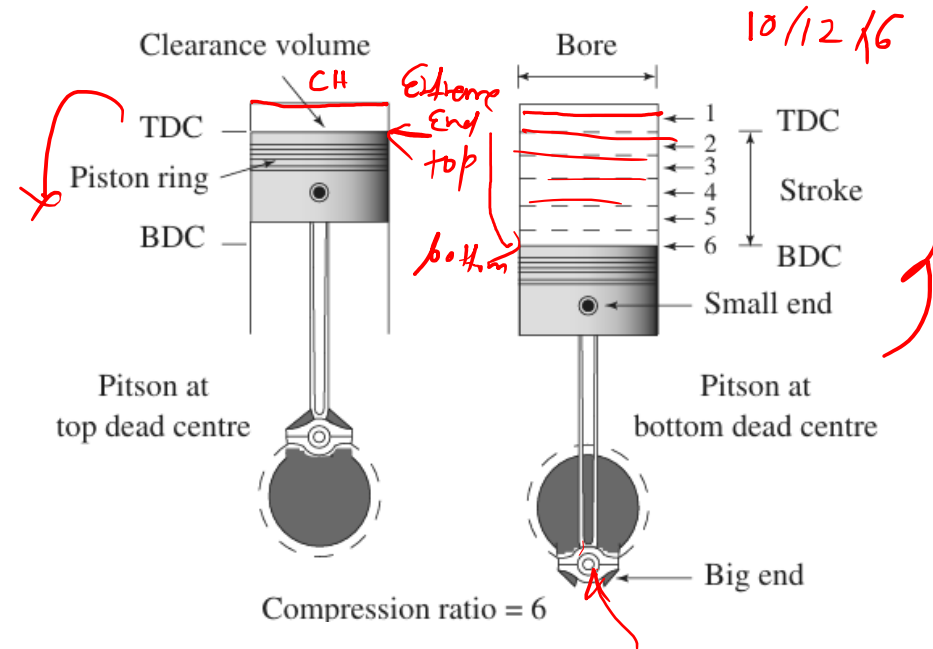
❖ Stroke-to-Bore Ratio (L/d)

- The **stroke-to-bore ratio (L/d)** is an important parameter for classifying engine geometry.
- If $d < L$, the engine is **under-square**; if $d = L$, it is **square**; and if $d > L$, it is **over-square**.

Dead Centre

The **dead center** is the position of the piston and its associated moving parts at the instant when the piston reverses its direction of motion at either end of the stroke. An engine has two dead centers:

- Top Dead Centre (TDC)**
- Bottom Dead Centre (BDC)**



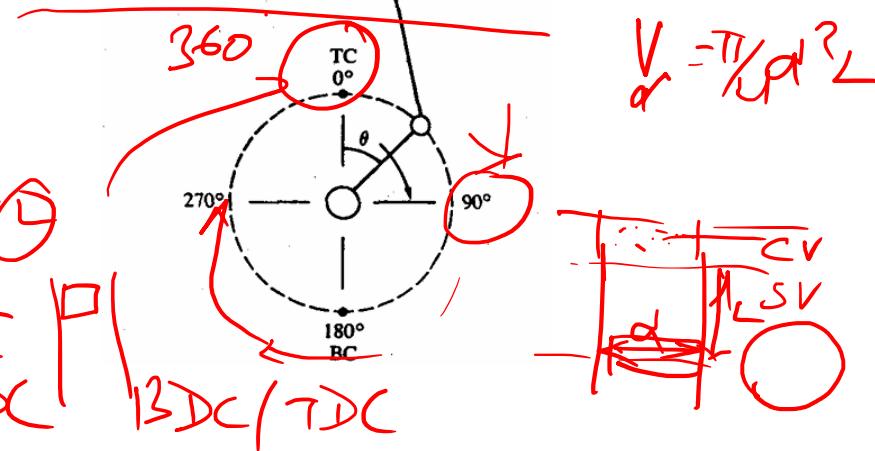
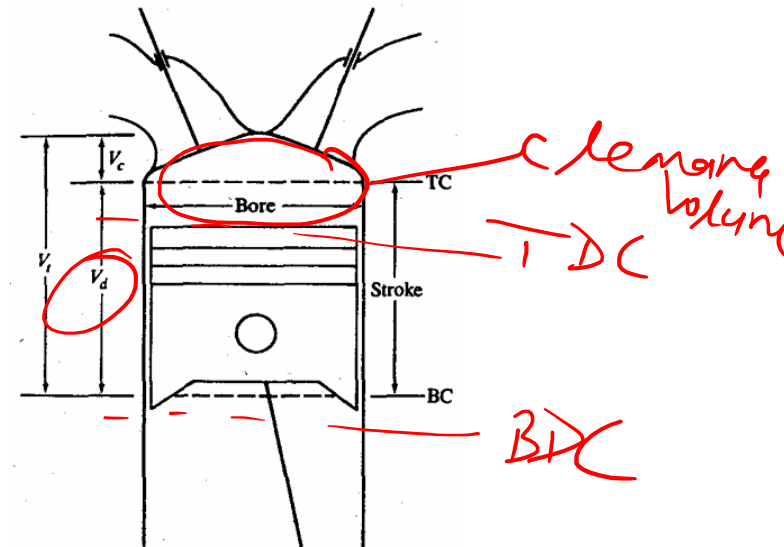
Nomenclature of SI Engine

Top Dead Centre (TDC): It is the dead center position at which the piston is farthest from the crankshaft. In vertical engines it is designated as **TDC**, while in horizontal engines it is referred to as **Inner Dead Centre (IDC)**.

Bottom Dead Centre (BDC): It is the dead center position at which the piston is nearest to the crankshaft. In vertical engines it is designated as **BDC**, while in horizontal engines it is referred to as **Outer Dead Centre (ODC)**.

Displacement or Swept Volume (V_s or V_d): It is the volume swept by the piston as it moves from one dead center to the other. It is expressed in **cubic centimeters (cc)** and is given by

$$V_s = A * L = \frac{\pi}{4} * d^2 * L$$



Handwritten calculations and notes:

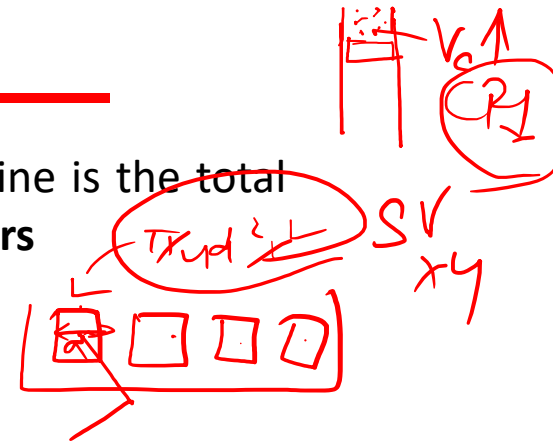
- 160 cc, 180 cc, 350 cc, 650 cc
- 1.24
- 110 cc
- cm³
- 13L 16L
- IDC (ODC)
- BDC / TDC

Nomenclature of SI Engine



Cubic Capacity or Engine Capacity: The **cubic capacity** of an engine is the total displacement volume of all cylinders. For an engine with **K cylinders**

$$\text{Engine capacity} = K * V_s \rightarrow$$



Clearance Volume (V_c): It is the **volume** of the combustion chamber above the piston when it is at **Top Dead Centre (TDC)**. It is expressed in **cubic centimeters (cc)**.

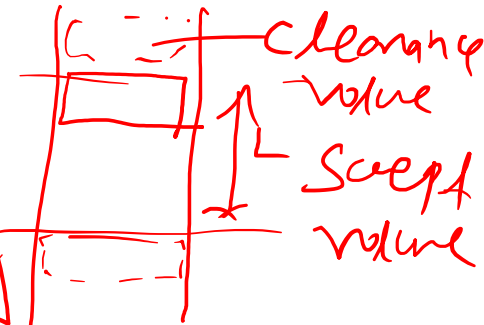


Compression Ratio (r): It is the ratio of the total cylinder volume (V_t) at Bottom Dead Centre (BDC) to the clearance volume (V_c). It is given by

$$r = \frac{V_t}{V_c} = \frac{V_s + V_c}{V_c} = 1 + \frac{V_s}{V_c}$$

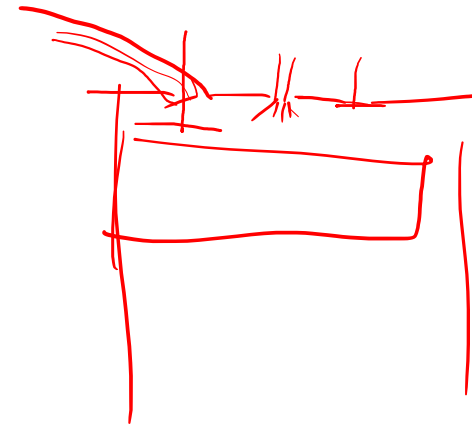
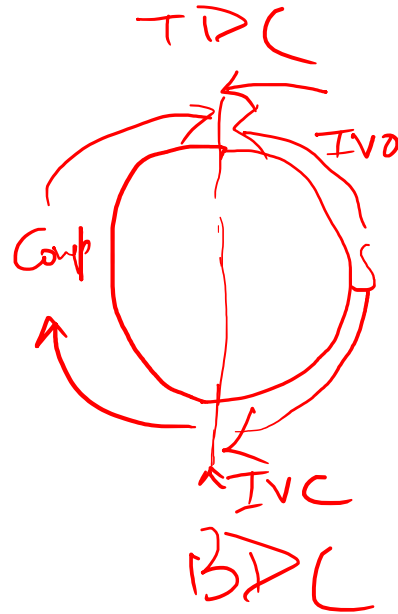
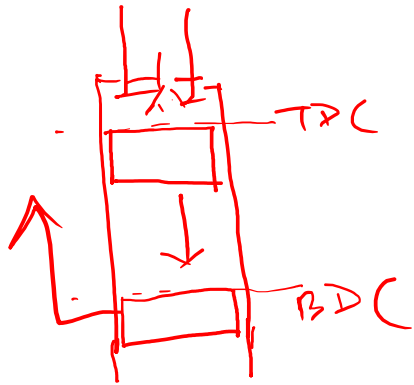
✓

$$CR = \frac{\text{Total volume} = V_c + V_s}{\text{min vol/c} = V_c} = 1 + \frac{V_s}{V_c}$$

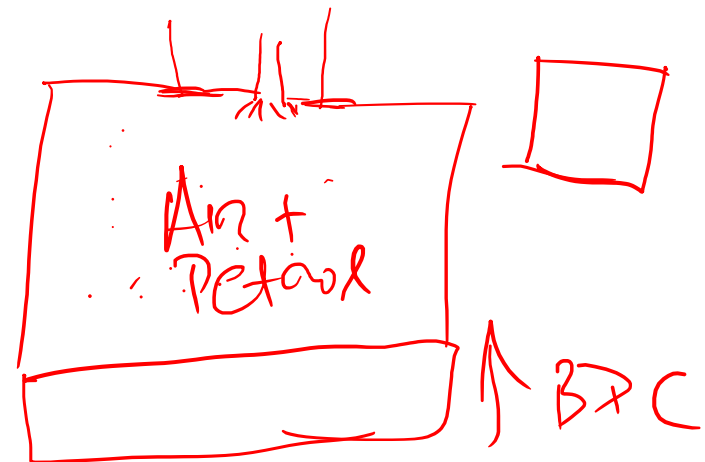
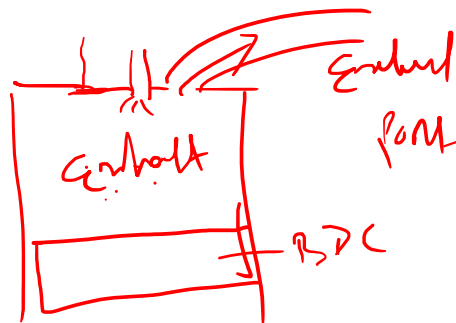
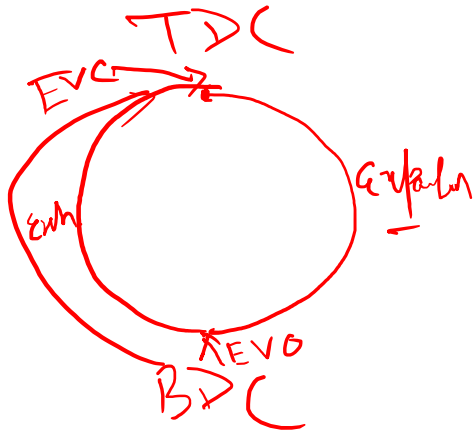


Advanced IC Engines

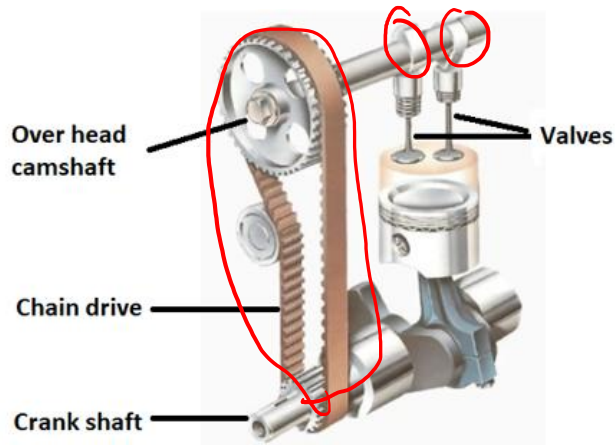
Valve Actuation



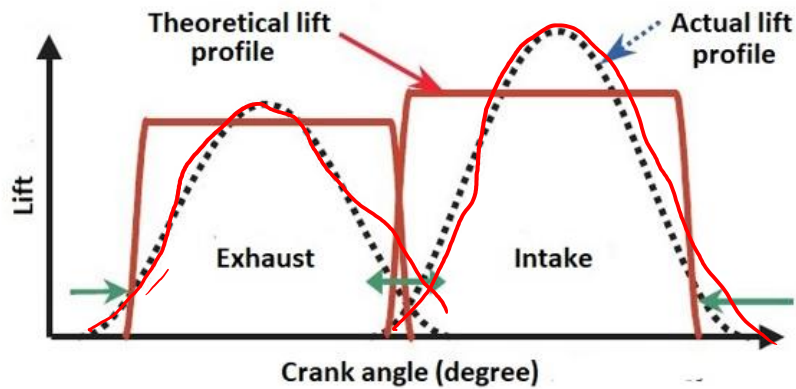
IVO
① Intake
valve
opening



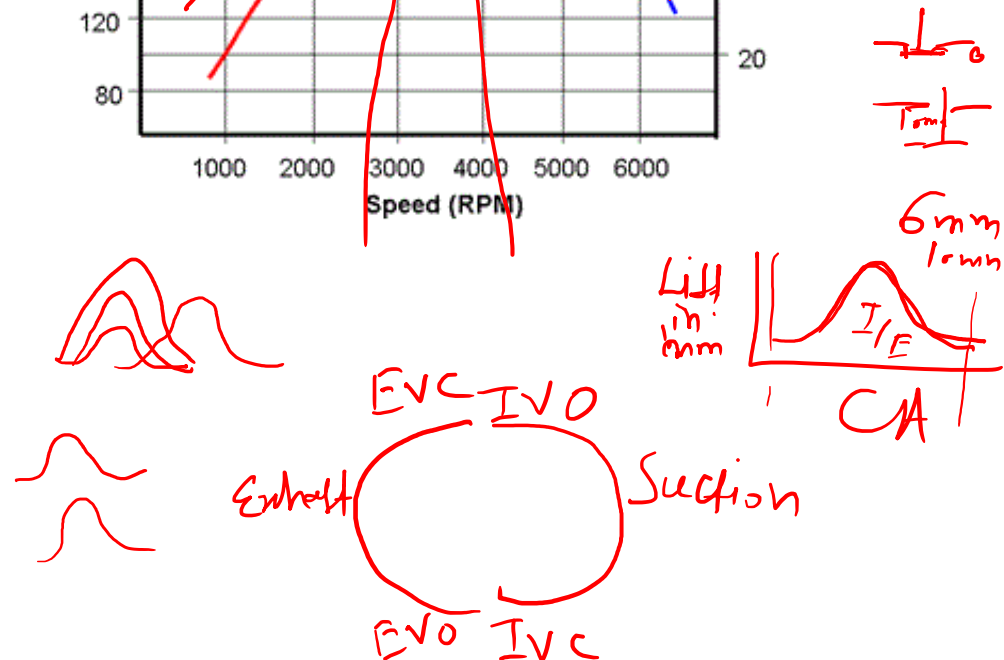
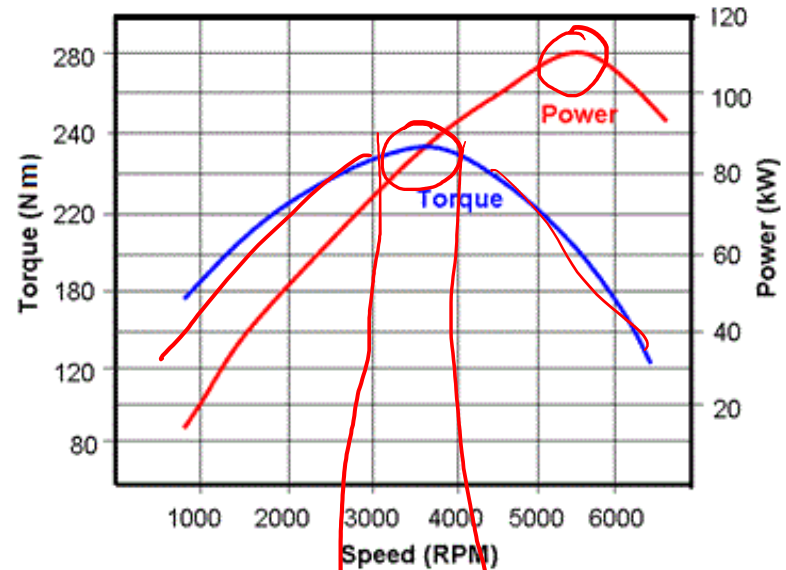
Valve Actuation



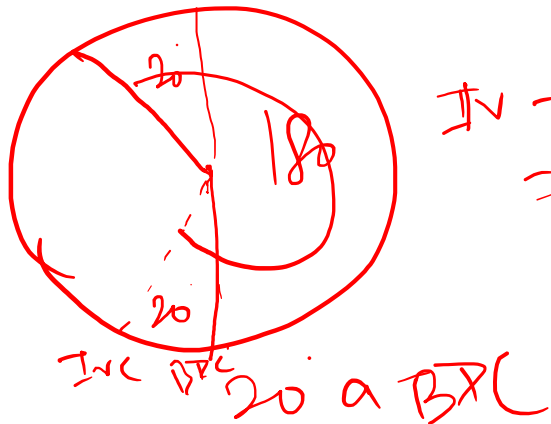
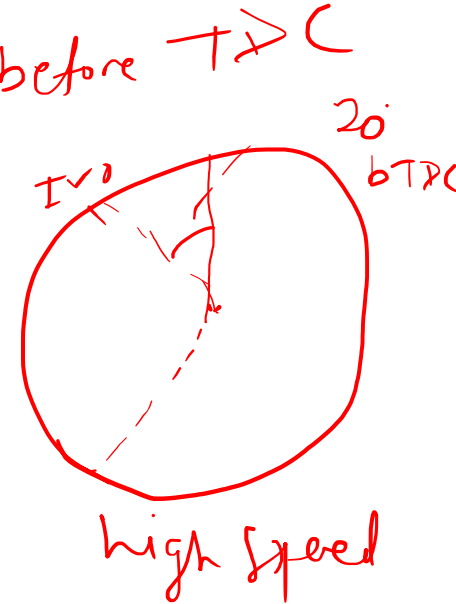
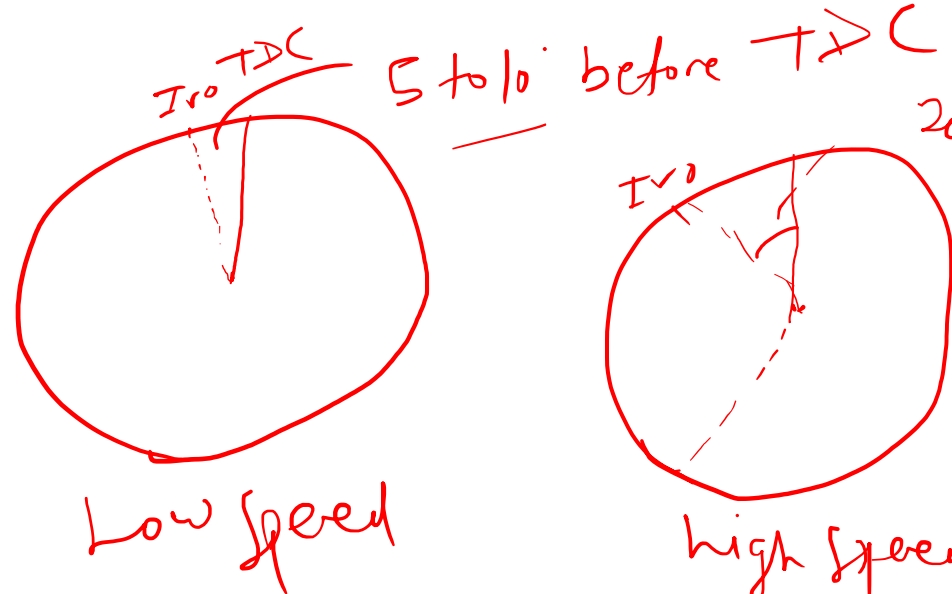
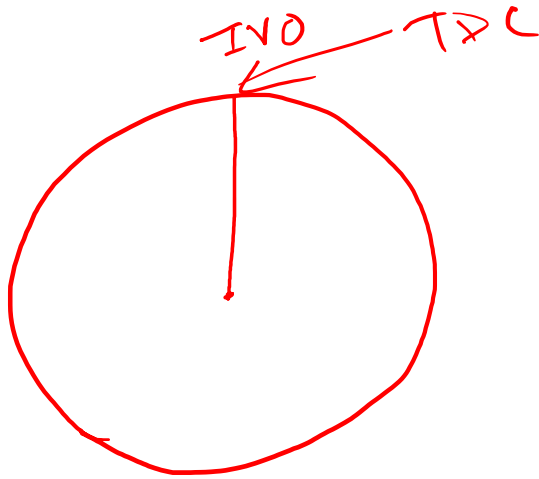
Valve actuation system of SI engine



Ideal Vs. actual valve lift profile



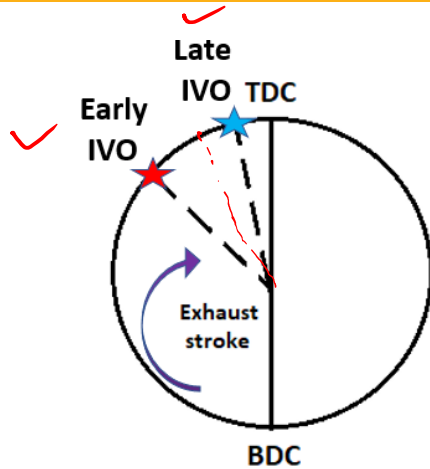
Valve Events



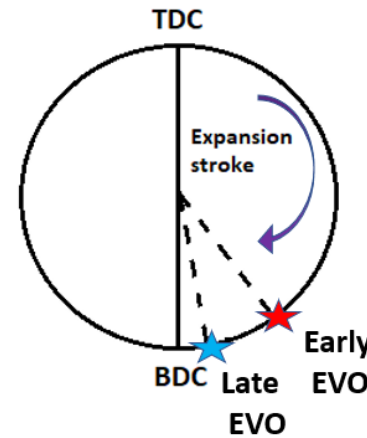
$$IV - 180 + 20 + 20 = 240$$



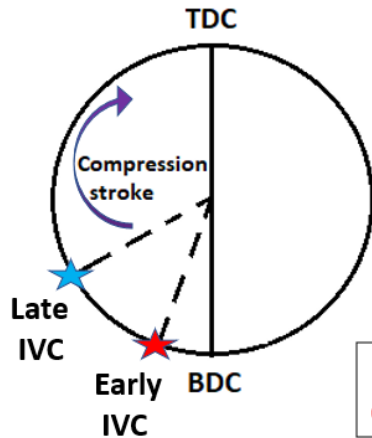
Valve Events



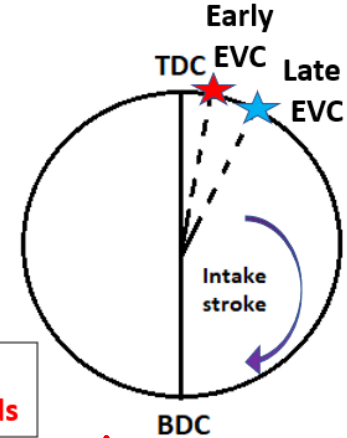
Intake valve opening (IVO)
 ✓ High speed-Early opening
 ✓ Low speed-Late opening



Exhaust valve opening (EVO)
 High speed-Early opening
 Low speed-Late opening



Intake valve closing (IVC)
 High speed-Late closing
 Low speed-Early closing



Exhaust valve closing (EVC)
 High speed-late closing
 Low speed-Early closing

Hence, the valve events need to be optimized for different engine speeds

& load condition

Variable valve actuation (VVA) system



❖ Variable valve actuation system

- In a VVA system, the intake and exhaust valve events are varied according to the engine operating conditions.

❖ Types

- Semi-flexible VVA system (Cam-based) ✓
- Fully-flexible VVA system (Cam-less) ✓

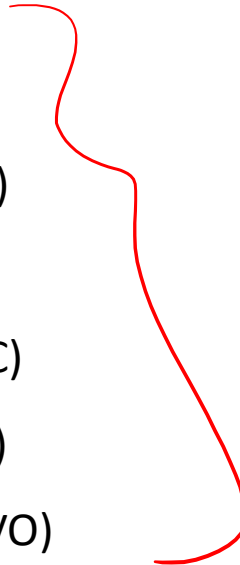


Variable valve actuation (VVA) system



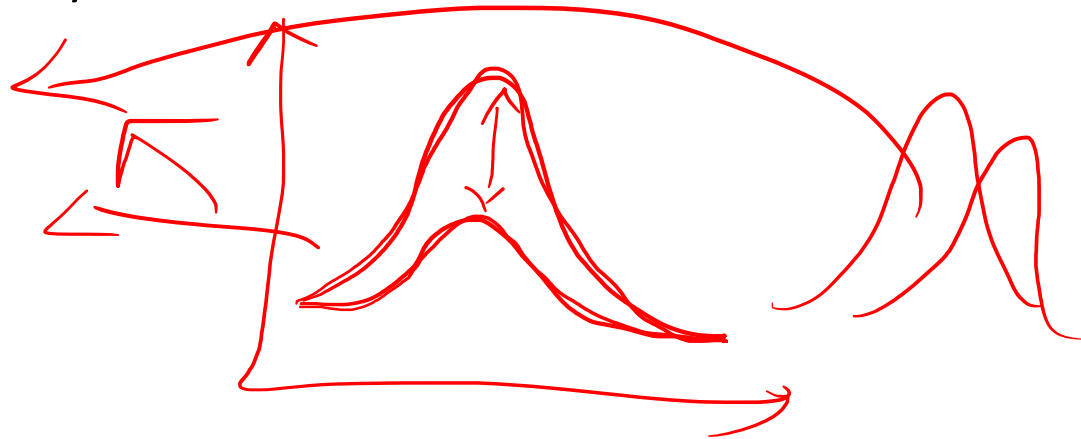
❖ There are eight ways the intake and exhaust valve timings are used.

1. Early intake valve closing (EIVC)
2. Late intake valve closing (LIVC)
3. Early intake valve opening (EIVO)
4. Late intake valve opening (LIVO)
5. Early exhaust valve closing (EEVC)
6. Late exhaust valve closing (LEVC)
7. Early exhaust valve opening (EEVO)
8. Late exhaust valve opening (LEVO)

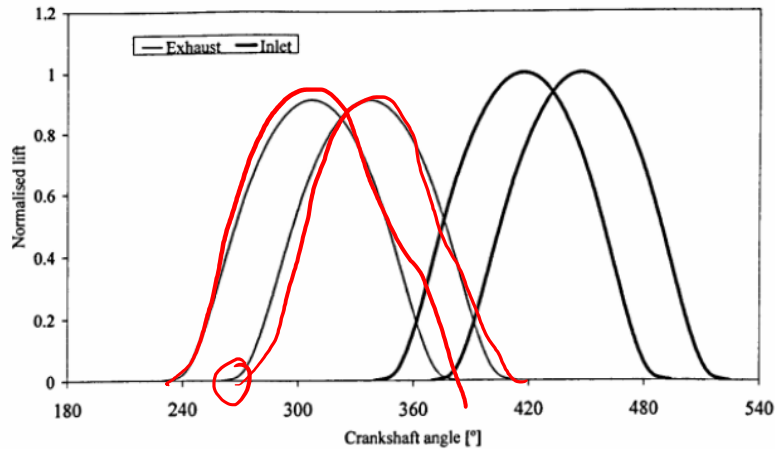


Semi-flexible VVA system

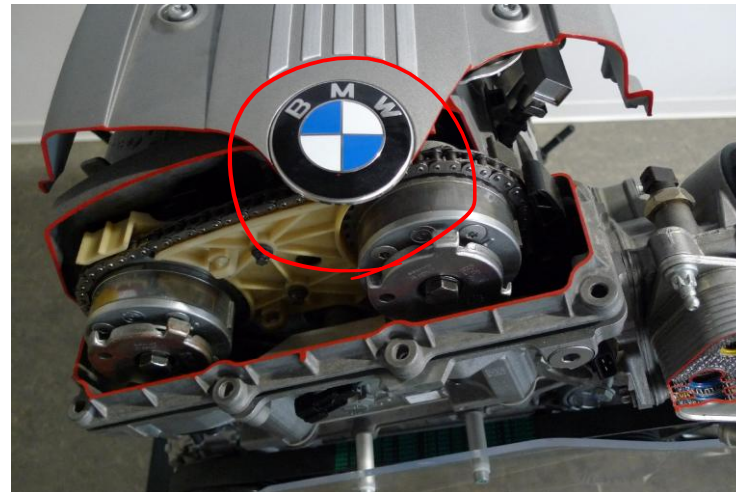
- ❖ In a semi-flexible VVA system, the cam mechanisms are modified, and the valve events are varied discretely.
- ❖ Current available semi-flexible VVA systems are:
 1. Cam phasing mechanism
 2. Cam switching mechanism



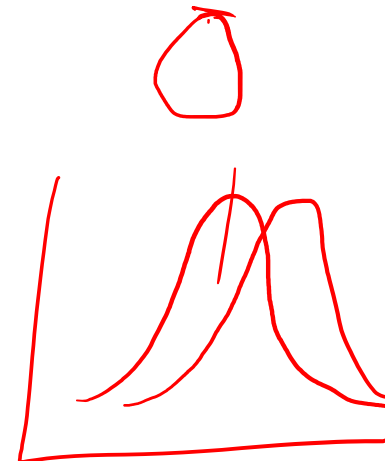
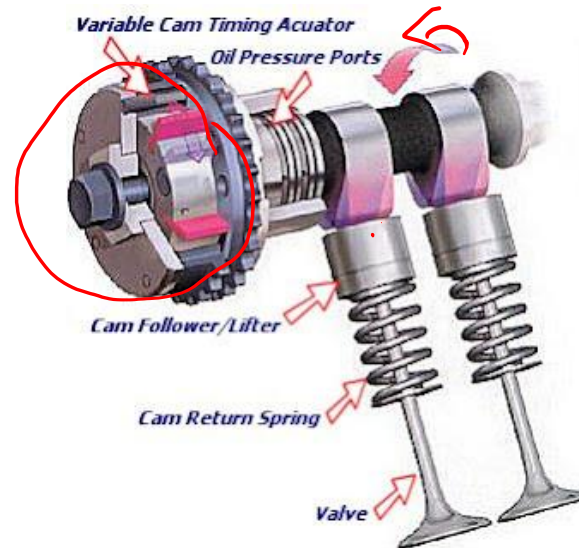
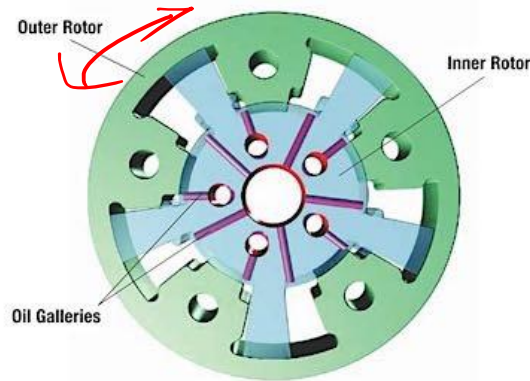
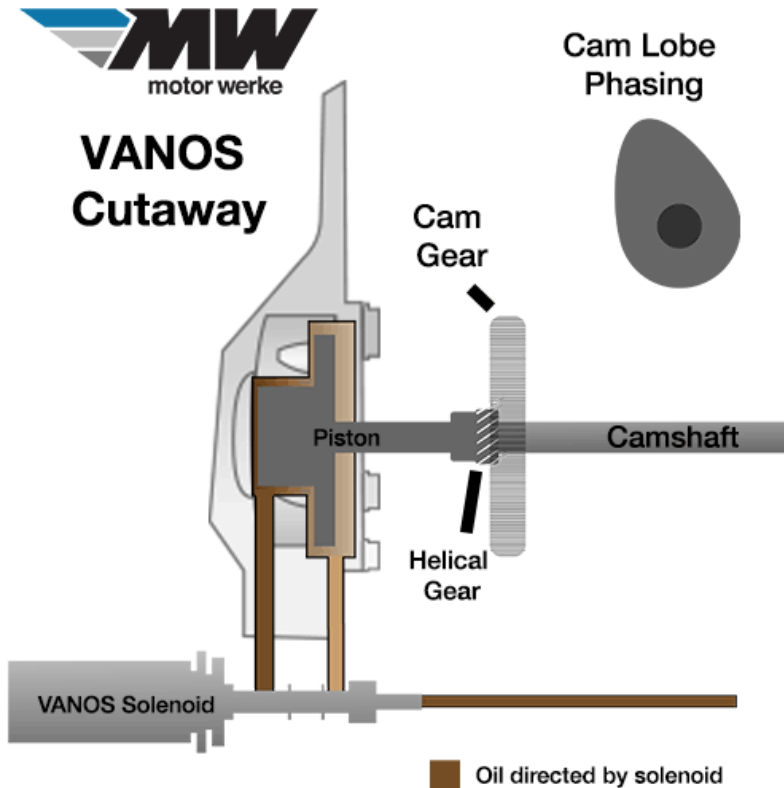
Cam phasing mechanism



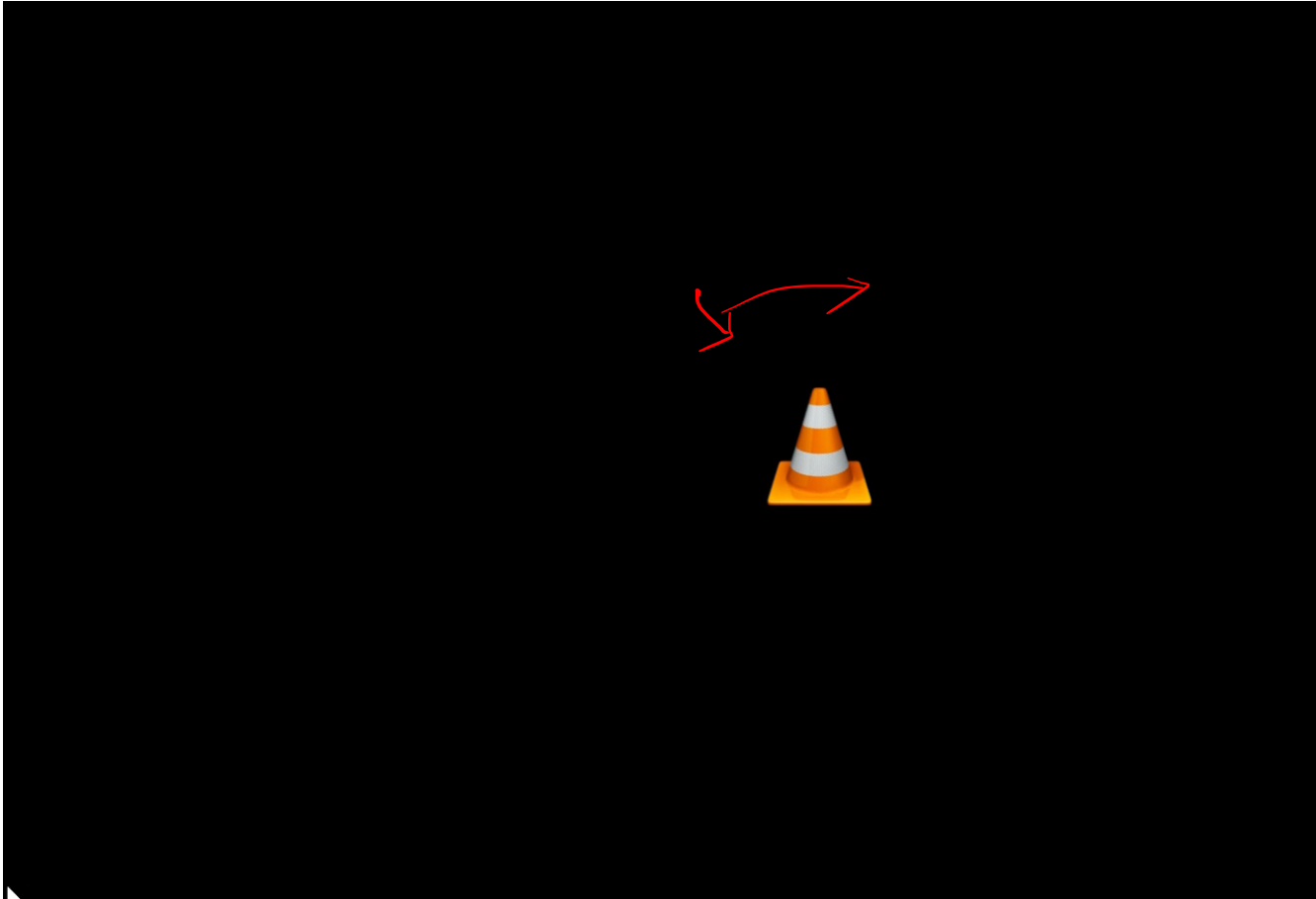
Phasing change / shift
angle change



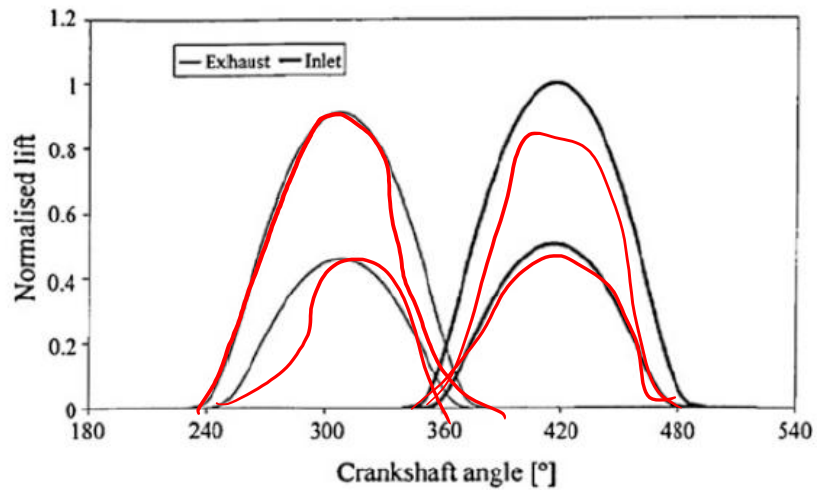
Cam phasing mechanism



Cam phasing mechanism



Cam switching mechanism



Cam switching mechanism



- ❖ Honda's VVT and Lift Electronic Control (VTEC) system uses a cam switching technique to vary the valve events



Cam switching mechanism

